

EfficientNAS: Progressive Search Space Pruning for Efficient Neural Architecture Search

Abstract

We introduce EfficientNAS, a neural architecture search framework for image classification that discovers architectures achieving 97.8% accuracy on CIFAR-100 and 89.2% top-1 accuracy on ImageNet. Unlike previous NAS methods requiring 3,000+ GPU hours, EfficientNAS completes the search in just 48 GPU hours using a novel progressive search space pruning strategy. The framework was implemented using TensorFlow 2.12 and trained on 8 NVIDIA V100 GPUs. Our discovered architecture, EfficientNAS-A1, achieves comparable accuracy to manually designed networks while using 40% fewer parameters. The key contribution is the Progressive Search Space Pruning (PSSP) algorithm that eliminates unpromising architecture candidates early in the search process.

1. Introduction

Neural Architecture Search (NAS) has emerged as a powerful paradigm for automating deep learning model design. However, the computational cost of NAS remains prohibitive for most researchers. The original NAS paper by Zoph and Le (2017) required 28,000 GPU hours to discover competitive architectures. Subsequent works like DARTS (Liu et al., 2019) reduced this to 4 GPU days but suffered from instability. Our EfficientNAS addresses both challenges. The main objective of this paper is to develop a computationally efficient NAS method that discovers high-accuracy architectures in under 50 GPU hours while maintaining search stability. We target image classification as the primary task, with transfer learning experiments on object detection.

2. Related Work

NAS research evolved through three main approaches. Reinforcement Learning-based NAS (Zoph and Le, 2017) used an LSTM controller but required 28,000 GPU hours. Evolutionary NAS (Real et al., 2019) employed genetic algorithms reducing cost to 3,150 GPU hours. Differentiable NAS like DARTS (Liu et al., 2019) made the search continuous, requiring only 1.5 GPU days but exhibiting performance collapse. Weight-sharing approaches like ENAS (Pham et al., 2018) achieved 1,000x speedup. Our PSSP method combines weight-sharing with progressive pruning, offering both speed and stability. Notably, transformer-based architectures have also been explored for NAS, with ViT-NAS (Chen, 2023) discovering efficient vision transformer configurations.

3. Methodology

EfficientNAS operates in three phases. Phase 1: Supernet Training - We construct a supernet containing all candidate operations (3x3 conv, 5x5 conv, dilated conv, separable conv, skip connection, max pooling, avg pooling, attention block). The

supernet is trained for 100 epochs using the full CIFAR-100 training set with SGD optimizer, learning rate 0.025, and cosine annealing schedule. Phase 2: Progressive Search Space Pruning (PSSP) - After every 20 epochs of supernet training, we evaluate all candidate operations at each layer using a validation subset. Operations performing below the median are pruned. This reduces the search space by approximately 50% each round, from 7^{14} initially to approximately 2^{14} after 4 rounds. Phase 3: Architecture Selection - From the pruned search space, we perform evolutionary search with a population size of 50 for 500 generations. Each candidate is evaluated using supernet weights without retraining. The top architecture is then retrained from scratch for 600 epochs. The framework uses TensorFlow 2.12 with custom Keras layers for the search space definition.

4. Dataset

Primary evaluation was performed on CIFAR-100, containing 60,000 32x32 color images across 100 classes (50,000 training, 10,000 test). We applied standard data augmentation: random cropping, horizontal flipping, and Cutout regularization. For large-scale evaluation, we used ImageNet (ILSVRC 2012) containing 1.28 million training images and 50,000 validation images across 1,000 classes at 224x224 resolution. The ImageNet experiments used RandAugment and MixUp augmentation. The total dataset size for CIFAR-100 training was approximately 150MB, while the ImageNet dataset required 150GB of storage.

5. Results

EfficientNAS-A1 achieved 97.8% accuracy on CIFAR-100 test set and 89.2% top-1 accuracy on ImageNet validation set. The discovered architecture contains 5.2 million parameters, which is 40% fewer than ResNet-152 (60M params) while achieving 1.3% higher accuracy. Compared to DARTS (97.2% CIFAR-100), we improve by 0.6% while using 30% less search time. The total search cost was 48 GPU hours on V100 GPUs, compared to 3,000+ hours for RL-based NAS and 36 GPU hours for DARTS (but with better stability). The architecture features 14 layers with predominantly 3x3 separable convolutions and attention blocks in the deeper layers. Inference latency was 2.1ms per image on V100 and 8.5ms on a consumer RTX 3080 GPU. The training cost for the final architecture retraining was approximately \$1,800 in cloud compute.

6. Transfer Learning Results

We transferred EfficientNAS-A1 to object detection using Faster R-CNN as the detection head. On COCO 2017 validation set, the model achieved 42.7 mAP, comparable to ResNeXt-101 backbone (43.0 mAP) but with 55% fewer FLOPs. Fine-tuning required only 12 epochs compared to 24 epochs for training from scratch. This demonstrates the strong transferability of NAS-discovered architectures.

7. Conclusion

EfficientNAS demonstrates that competitive architectures can be discovered in 48 GPU hours through progressive search space pruning. The discovered EfficientNAS-A1 achieves 97.8% on CIFAR-100 and 89.2% on ImageNet with 40% fewer parameters than comparable manually-designed networks. Future work will explore multi-objective NAS optimizing for accuracy, latency, and energy consumption simultaneously.